

RIPAN BAIDYA

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Work Experience

Orbitdynamix

Apr 2025 – May 2026

Java Developer

Kolkata, India

- Reduced semantic query latency by **85%** (1,100ms to 160ms) and cut LLM API costs by **40%** by architecting a per-user **Redis** cache with **PgVector HNSW** cosine similarity search.
- Built **80+** high-performance REST APIs for a mobile application, developing unit and integration tests using **JUnit** and **Mockito** to improve API reliability.
- Built an instant messaging and invoicing feature, leveraging **WebSockets** for real-time notification and implementing **Thymeleaf** to automate email invoice dispatch.
- Built a secure shopping cart and checkout system, utilizing **Redis** to cache active shopping items and integrating the **Razorpay** SDK for payment processing and error handling.
- Automated administrative tasks for a gym backend serving **500+ active users**, eliminating **20+ hours** of manual effort weekly.

SHA Infotech Solution Pvt Ltd

Jun 2024 – Sep 2024

Java Developer Intern

Kolkata, India

- Engineered a secure backend for a student management system using **Spring Boot** and **MySQL**, centralizing attendance and academic workflows for **200+ students**.
- Accelerated data retrieval by **25%** through query optimization and indexing of student, attendance, and grade tables in **MySQL**.

Projects

LuneCare: Distributed Doctor Appointment Booking Platform

[GitHub](#) | [Live](#)

- Resolved concurrent booking conflicts by architecting a choreographed **SAGA pattern** using **RabbitMQ** across distributed microservices.
- Secured microservice endpoints by centralizing **JWT** authentication and rate limiting at the API Gateway using **Spring Cloud Gateway**.
- Prevented cascading failures during outages by implementing **Resilience4j** circuit breakers and retry with backoff mechanisms.
- Accelerated troubleshooting cycles by architecting a unified observability pipeline using **Prometheus**, **Grafana**, and **Loki** for metrics and log aggregation.
- Ensured deployment consistency and system scalability by containerizing microservices with **Docker** and orchestrating them via a **Kubernetes** cluster.

WalletIQ: RAG-Based AI Expense Tracker

[GitHub](#) | [Live](#) | [Video](#)

- Engineered a user-scoped **RAG pipeline** using per-user **Redis** semantic search to dynamically ground LLM responses in transaction, budget, and savings data.
- Built an interactive analytics dashboard utilizing **WebSockets** for real-time notifications and dynamic charts for visual financial tracking.
- Automated the deployment of **Docker** images to **GCP Compute Engine** via **GitHub Actions**.

Education

Dinabandhu Andrews Institute of Technology & Management

Sep 2022 - Jun 2025

Bachelor of Computer Application

CGPA: 8.93/10

Technical Skills

Languages: Java, Python, SQL

Backend: Spring Boot, Spring AI, REST APIs, Microservices, Hibernate/JPA, JUnit, RabbitMQ

Frontend: TypeScript, React.js, Tailwind CSS

Databases: PostgreSQL, MongoDB, Redis

Cloud & DevOps: GCP (Compute Engine, GKE, Cloud Storage), Docker, Kubernetes, Git

Achievements

- Guardian** on LeetCode with highest **rating 2290 (Top 1% globally)**.
- Secured global ranks **73**, **149**, and **221** in **LeetCode Weekly/Biweekly Contests 465**, **164**, and **468** respectively.
- Secured **2nd Place** out of **40+ teams** at the **TechQuest** Coding Competition.